

**PROJECT HIMSOG (HEALTHCARE INFORMATION AND
MEDICAL-SERVICES SEARCH WITH ONLINE-BOOKING AND
GEOLOCATION): ENHANCING HEALTHCARE ACCESSIBILITY AND
INFORMATION MANAGEMENT FOR DIGOS CITY**

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In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology

by

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Researchers

DEDICATION

The researchers dedicate this capstone project to all individuals and communities who continue to face challenges in accessing reliable and timely healthcare information. This project is offered as a contribution toward improving healthcare accessibility and informed decision-making for the residents of Digos City.

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ABSTRACT

Access to reliable and centralized healthcare information remains a challenge due to fragmented data sources, limited digital health literacy, and uneven access to services. In Digos City, residents often encounter difficulties in locating appropriate healthcare facilities and services in a timely manner. This capstone project presents Project HIMSOG (Healthcare Information and Medical-Services Search with Online-booking and Geolocation), a hybrid web-based application designed to improve healthcare accessibility and information management within the city. Project HIMSOG consolidates healthcare information from clinics, veterinary services, and government health programs into a single platform. The system utilizes geolocation and mapping technologies, supported by the K-Nearest Neighbors (KNN) algorithm, to identify nearby healthcare providers based on user location. The system also features online appointment booking, service filtering, and administrative analytics to monitor usage and service demand. Developed using modern web technologies, the platform is accessible through web browsers and Android devices. The implementation of Project HIMSOG demonstrates effectiveness in enhancing healthcare service discovery, supporting informed decision-making, and promoting efficient access to healthcare services for the residents of Digos City.

Keywords: *Healthcare Information System, Geolocation, Online Booking, Digital Health, Web Application*

SDG: *3 Good Health and Well-being; 9 Industry, Innovation and Infrastructure*

EXECUTIVE SUMMARY

The availability of reliable healthcare-related information in Digos City continues to be a challenge for residents, partly because of numerous available options and limited access to effective means to book appointments. To address this, Project HIMSOG was developed as a hybrid web-based solution to consolidate information about healthcare providers and services into one location. Project HIMSOG also utilizes geolocation capabilities to direct users to their closest healthcare providers and allows users to reserve appointments online, and filter by type of service. The system also provides administrators with basic analytics to assess the level of demand of services. As a result, Project HIMSOG is a viable digital solution for ensuring residents of Digos City have greater access to necessary health-related services, improved healthcare accessibility, patient convenience, and coordination among Digos City's local healthcare providers.

Categories and Subject Descriptors

CCS → Information systems → Geographic information systems → Health information systems

General Terms

Healthcare Information Systems, Geolocation, System Design, Algorithms, Human Factors, Data Management, Performance, Security

Keywords

Project HIMSOG, Healthcare Information System, Online Appointment Booking, Geolocation Services, K-Nearest Neighbors (KNN), Location-Based Recommendation, Web Application, Mobile Application, Digital Health, Healthcare Accessibility, Digos City

1. INTRODUCTION

1.1 Project Context

One of the pillars of Sustainable Development Goals (SDGs) is Good health and Well-being which aims to ensure healthy lives and bring well-being to all people regardless of ages and genders [1]. One of the main challenges in achieving SDG 3 is the inequality in access to healthcare [2]. This includes people's ability to search and access scattered information from e-sources, reviews and use the gathered information for their health-related decision-making. People need skills and knowledge to navigate through these different information platforms to process and filter information to make health decisions [3].

Health Literacy is defined as “the degree to

which individuals have the ability to find, understand, and use information and services to inform health-related decisions and actions for themselves and others” [4]. Digital health literacy (DHL) has the same definition but in the context of technology [5]. DHL varies on individuals' literacy level, socioeconomic background, age and familiarity with technology, also the motivation to seek information online [3]. Traditionally, health dissemination is primarily through direct communication between healthcare providers and patients. Gradually, it transitioned to patients obtaining and searching information through Google searches and social media platforms [6]. According to one study, 11 articles mentioned that older adults' use social media platforms like Facebook, Twitter and Blogs to seek online health information [7]. The information they gather online is also used for cross-referencing on the information they gather elsewhere. As social media being one of the powerful mediums for health related information, people now face information overload, validity of their findings, misinformation, disinformation and often contradictory information which can cause confusion and inaccuracy in health-related decision-making [8, 9]. Another study mentioned that health information on the internet took an average of 4 searches to find relevant topics within mass media which suggests the inefficiency in finding health information (FHI) [9].

Turning to another aspect, the US healthcare system is shifting towards processes focused on patients and increased transparency. The study defines transparency as the availability and accessibility of information to the public [10]. In the 2019 Consumer Health Insights Survey, merely a small percentage of participants reported successfully obtaining the information they needed for healthcare decisions, leading to low satisfactory levels. There are some disparities between healthcare information they seek, what is accessible, and its location [11]. Apart from the lack of transparency, it has been shown that many individuals face challenges due to limited capacity to access healthcare services, primarily because of a lack of awareness regarding the available services, their location, and the methods to reach them [12]. This issue is further compounded by geographic disparities, as spatial accessibility to patients in the Philippines remains uneven and inadequate [13]. In response to these challenges, Project HIMSOG aims to provide a digital central

information platform for providing all the necessary healthcare information in one accessible location. The system contains searchable directories of clinics, veterinary services, operation hours and the insurance providers that they are accepting and integrated mapping and geolocation services to help users find the nearest and most appropriate providers. The system also allows appointment booking, present price ranges for various services, and filters to sort providers by location, specialty, and availability. Altogether, these functions are in a single user-friendly platform that aims to improve access to healthcare, enable proper service utilization, and equip the constituents of Digos City to make informed and timely health decisions.

Several digital health platforms have come up for addressing access issues and coordination in healthcare and foremost among them are Project Clinic, Zocdoc, and Doctolib. Firstly, Project Clinic is a multi-platform web-based application that facilitates making appointments. The system is aimed at providing accessibility and ease to the patients as well as the administrators of this facility-the application is chosen because it runs smoothly on laptops, tablets, and smartphones.. [14]. Zocdoc on the other hand is an online healthcare platform made particularly for finding primary care doctors, and booking appointments. They tend to take added advantage of clarity and informed decision-making, as it would show the satisfaction score of patients and average waiting times in addition to specific physician characteristics such as education, language, and even facial attributes [15]. Lastly, Doctolib is an AI-augmented and intelligent healthcare platform that offers patients a way to book appointments in real-time, predicts events, provides customer assistance at any time by chatbots, and has telemedicine services. The aim is operational efficiency and patient involvement,[16].

All those systems offer access to online booking (Project Clinic), decision support systems through reviews (Zocdoc), and teleconsultation powered by AI and efficiency (Doctolib), among others. However, these systems primarily seem to be designed for broad, commonly urban, or technologically advanced contexts, and therefore most likely cater to the country-specific geographic, socioeconomic, and digital literacy extents of small cities or rural areas such as Digos City. Usually, these modalities favor facilities run by

medical clinics but do not include features for veterinary services or even free government health programs or localized content tailored to the healthcare system in the Philippines. The distinguishing feature of Project HIMSOG is its holistic and local context design. The system is not only an appointment and geolocation android web application, but also brings together data from clinics, veterinary services, and free government programs onto one platform. The system boasts real-time updates, accepted insurance, filtering by specialty and availability, and instant messaging for patient and provider communication. Project HIMSOG's emphasis on accessibility and transparency for the residents of Digos City makes it truly different compared to those reviewed systems. The platform also considers the local context concerning fragmented healthcare data, challenges of digital health literacy, and spatial differences in access to services, thus giving users the power to make better health decisions that are well-timed, relevant, and informed.

LEGEND :

- A. Project HIMSOG
- B. Zocdoc
- C. Doctolib
- D. Project Clinic

Table 1: Features Comparison of Different Relevant System

FEATURE	A	B	C	D
Centralized Directory	✓	✗	✗	✗
Web and Mobile Integrated System	✓	✗	✗	✗
Geolocation and Mapping Integration	✓	✗	✗	✗
Real-time Operating Hours and Availability	✓	✗	✓	✗
Local Context Focus	✓	✗	✗	✗
Search Filters (e.g., specialty)	✓	✗	✓	✗
Appointment Booking	✓	✓	✓	✓
Inclusion of Government Health Services	✓	✗	✗	✗

1.2 Purpose and Description

1.2.1 Purpose

The purpose of this system is to change how residents of Digos City find, access, and manage health services. The objectives support the Sustainable Development Goals (SDGs) directly, with SDG 3, that is, Good Health and Well-being, being particularly relevant, as it guarantees access to relevant health services. With that, the system also acts as a portal in bringing together various medical resources offered by the community through the aggregation of locally dispersed sources of data-and in that sense, satisfy SDG 9: Industry, innovation, and infrastructure. Project HIMSOG embodies the modernization of web technologies and data integration methods to make healthcare decisions and promote improved health outcomes for the good health and well-being of the community [1][19].

1.2.2 Description

Project HIMSOG is a web and mobile application aimed at enabling residents of Digos City to easily access, locate, and connect with healthcare providers. In essence, this android web platform is a repository from which medical-related information can be searched online: That is, users can search for clinics, veterinary services, and free government health programs in the immediate vicinity. Using geolocation and mapping capabilities, users can find a healthcare facility according to their location. Furthermore, the system provides details of private clinics, veterinary clinics, and other specialized clinics also the insurances that they are accepting, unavailability of service, and the contact number for inquiries. To further improve the user experience and access to healthcare, Project Himsog allows online booking of appointments and filtering for services.

1.3 Objectives of the Study

To develop a hybrid and web-accessible application with location-integrated feature for Digos City enabling residents to get in-depth and up-to-date information about health services for informed decision-making, service utilization, and health outcomes.

- 1.3.1 To design and implement a digital platform that integrates and consolidates all essential healthcare information such as clinic services, free barangay health services, operating hours, accepted insurance, and contact details, into a single portal for easy access for the city of Digos residents.

- 1.3.2 To utilize geolocation and mapping technologies, integrated with the K-Nearest Neighbor (KNN) algorithm, to enable users to conveniently identify and navigate to nearby healthcare providers.

- 1.3.3 To integrate descriptive data analytics that enables administrators to monitor platform usage with respect to booking history (daily, monthly) and often searched services thus deriving insights into trends in healthcare services, demand for service providers, which facilitates data-driven planning, resource allocation, and strategy formulation.

1.4 Scope and Limitations

1.4.1 Scope

Project Himsog scope includes the design of a web-based application that host a searchable list of health service providers, namely, private clinics, free government health centers, and veterinary services with other pertinent information about hours of operation, contact information, insurance accepted, list of services, and price ranges information. Utilizing geolocation and interactive mapping to help users find the nearest clinic to their location. The system allows appointment booking and service filtering based on location and specialty.

1.4.2 Limitations

The project's limitations with the proposed system is that it may not reach older people who have lower socioeconomic status and have low digital literacy and rely mostly on printed and traditional media (printed material, news paper, radio, television) for health related information [17]. The system is also solely for Digos City and is not applicable for other regions. Since the platform is internet-dependent, areas in Digos with poor internet connection do experience difficulties in accessing its full features.

Project Himsog is constrained by the way it currently authenticates healthcare providers . At the present moment, the super administrator is doing the verification by manual document review. Although this method guarantees accuracy and adherence to local laws and regulations, it may also lead to longer waiting times for approval, as well as increasing the workload of the super administrator, especially

with the continuous rise in the number of provider registrations.

1.5 Ethical Considerations

The implementation of Project HIMSOG abides by a robust series of ethical safeguards aimed at ensuring that the rights, privacy, and welfare of all involved individuals, healthcare providers, and institutions are duly respected and protected. This include the following ethical considerations:

1.5.1 Privacy and Confidentiality

All personal and health information obtained from patients, health care providers, and facilities is kept confidential. Identification of patients is, however, restricted to authorized personnel who ensure that all reasonable measures are implemented to protect the anonymity of sensitive information.

1.5.2 Informed Consent Process

All health practitioners and patients receive full disclosure regarding the aim, scope, and nature of the data collection and the intended use of such data on the platform. Consent was sought prior to any data collection, thereby allowing for voluntary participation predicated on informed understanding of the project.

1.5.3 Risks

Risks are identified and minimized with tight security, regular monitoring, and user education. This includes data breach, misinterpretation of information, or misappropriation of the platform features. Mitigation strategies are developed to solve ethical or technical issues if any.

1.5.4 Conflict of Interest

The project is open so that there is no conflict of interest. No provider or institution is given any undue advantage or promotion in the platform on criteria that are not justified and openly disclosed. Any links or partnerships are declared publicly.

1.5.5 Data Security and Integrity

The project ensures access protection, data loss prevention, and data modification through encryption, secured servers, and frequent data backup. Data validation protocols are instituted to measure the accuracy and reliability of stored information.

1.5.6 Research Integrity

Principles anchored in truthfulness, openness, and accountability govern the researchers along the project lifecycle. Thus there can be no data fabrication, data manipulation, or

selective reporting, and all sources of information are recognized.

1.5.7 Non-Discrimination and Accessibility

The platform ensures that the health information is made accessible to all residents of Digos City, regardless of gender, age, or socio-economic status, or perhaps disability status. Efforts would be geared towards including the user-friendliness and accessibility of the platform for individuals with poor digital literacy.

II. METHODOLOGY

2.1 Data Gathering

This phase involved gathering of both primary and secondary data in a systematic way for Project Himsog to be developed fully. The secondary data was mainly getting the preliminary information regarding clinics, and other healthcare providers available in Digos City from public sources, such as websites and social media pages. It helped us make a sample list of the providers with their addresses and services. Afterward, primary data collection had to take place by means of on-site visits to check the accuracy of details like operating hours or addresses for geolocation and mapping features. The insights driven from both approaches proved to become very important in the system's design and functionality.

2.2 Requirement Analysis

2.2.1 User Requirement

The Project HIMSOG is a hybrid and web-accessible application that centralizes access to healthcare services and information for residents of Digos City. To efficiently secure the platform's functionality, myriad users are classified according to their roles vis-a-vis their respective responsibilities and areas of access. Several user roles with designated responsibilities and permissions ensure security, organization, and efficiency with respect to the operation across all service areas. These user roles include:

2.2.1.1. Administrator

2.2.1.1.1 System Administrator

The platform manager, responsible for viewing all user activity and the integrity of the system, is the System Administrator. It is his/her role to approve and verify accounts of healthcare providers, monitor analytics and usage of the platform, moderate content generated by users such as reviews, and ensure adherence to data privacy policies. The System Administrator further carries out the management of integrity

on the database and performs technical updates.

2.2.1.2 End User

2.2.1.2.1 General User/ Resident

This role represents the primary users of the system: those residing in Digos City and looking for healthcare services. These users actively engage in searching for clinics, veterinary services, and free government programs through the platform. They have the ability to filter services according to service type, area, availability, and insurance. They can also make an appointment, raise inquiries with providers, and give feedback on the service.

2.2.1.2.2 Healthcare Provider

This refers to individual clinics, veterinarians, and free government health service units that play a role in public health activities. Providers manage their facility information on the platform, respond to queries, book and organize appointments, and maintain the operational information like hours, rates, and types of accepted insurance.

2.2.2 Hardware and Software Requirements

In order to guarantee the best performance, usability and compatibility of the system on the widest variety of devices, the following hardware and software requirements are recommended. These requirements enable the system to meet all necessary conditions for seamless data submission and real-time geolocation-enabled interaction between end users (health seekers) and administrators (health facility managers, evaluators).

Table 1:Hardware Requirements

Hardware Requirements	
Hardware	Requirement
Processor	Intel Core i3 or AMD Ryzen 3 or Higher
Storage	256GB or Higher
RAM	8GB or Higher
HDD/SDD	30Gb (32bit); 50GB (64bit)
Mobile Device	Android 8.0 (Oreo) or latest

Table 2:Software Requirements

Software Requirements	
Software	Requirement
Operating System	Windows 10 or Higher
OS Version	Windows 10 or above version
Browser	Google Chrome or Microsoft Edge

2.2.3 Functional Requirement

The end-users of the system are primarily of three types namely:1. General User/Resident 2. Healthcare Provider 3. System Administrator

The distinct functionalities and the roles assigned to each of these user types are described in the further subsections.

2.2.3.1 General User / Resident

2.2.3.1.1 Registration and Profile Module

A new user is enabled to make a personal account through this module. The whole process includes: (1) completion of a registration form bearing basic information like full name, e-mail address, contact number and password; (2) optionally adding the user's address for location-based features; (3) users must validate their email address by clicking the link sent to them to complete their account and all of its features.

2.2.3.1.2 Module for Login

This module allows registered users to experience secure access to the accounts through their email and password. Users can also reset the password following instructions sent to the verified email address by the "Forgot Password" feature.

2.2.3.1.3Main Dashboard/ Search Module

The primary user interface serves as the main starting point for searching healthcare services within the system. It provides a centralized area where users can explore available specialized services. At the core of this interface is a search bar that allows users to enter keywords related to services, clinics. Alongside this, the interface includes an integrated filter and sorting mechanism that enables users to refine their search based on their specific needs. These parameters include the type of service—such as clinics, or veterinary centers—medical specialties like Dentistry or General Medicine, location or

proximity, operating hours, and even accepted insurance types. Together, these features ensure that users can efficiently navigate the platform and locate the most relevant healthcare options.

2.2.3.1.4 Geolocation and Mapping Module

This module helps users locate and identify nearby providers. It shows a map that displays the location of clinics based on their current location or a provided address, and provides navigation options.

2.2.3.1.5 Healthcare Provider Profile Module

This module allows users to view detailed profiles of healthcare providers. Information provided with the facility's name, address, contact details, a list of services offered with price ranges, operating hours, photos of the facility, and accepted insurance plans.

2.2.3.1.6 Online Appointment Booking Module

This is a platform, helping the users on how to set an appointment. The system help users to: 1. Select a service, date, and time from available slots set by the provider. 2. Confirmed appointments are sent through email and within the app itself. 3. Users can also keep track of upcoming and past appointments.

2.2.3.1.7 Feedback and Review Module

Users are allowed to rate and review their experiences with a provider on this module after the appointment. Such reviews can be made visible on the profile of the provider by choice to assist other users in making informed decisions.

2.2.3.2 Healthcare Provider

2.2.3.2.1 Business Registration Module

This module allows the facilities (i.e. clinics, vets, etc.) to register their businesses on the platform. The process involves: (1) Filling up a form with business related details, including facility name, address, contact information, and business license number. (2) Creating an administrator account for the facility. (3) Submitting necessary documents for verification by the System Administrator to become a trusted provider.

2.2.3.2.2 Provider Dashboard Module

This is the main interface for healthcare providers to manage their presence on the platform. It provides access to all management features, including appointment calendars, and profile settings.

2.2.3.2.3 Profile and Service Management Module

This module allows providers to create and update their public profile. They can: (1) List and describe the services they offer. (2) Set their operating hours and availability. (3) Update contact information and accepted insurance providers. (4) Upload photos of their facility and staff.

2.2.3.2.4 Appointment Management Module

This module enables providers to manage incoming appointment requests. They can: (1) View their upcoming schedule of appointments. (2) Confirm or decline new booking requests from users. (3) Mark appointments as completed.

2.2.3.3 System Administrator

2.2.3.3.1 Secure Login Module

This module allows the administrator to log in to the system's backend using secure credentials.

2.2.3.3.2 Administrator Dashboard

This module helps the System Administrator with an organized overview of all platform activity, including the number of registered users, active healthcare providers, recent bookings, and system health status.

2.2.3.3.3 Provider Verification Module

This module allows the administrator to review registration submissions from new healthcare providers. The administrators are responsible for verifying the authenticity of submitted documents and approving their accounts to ensure all listed providers are legitimate.

2.2.3.3.4 User and Content Management Module

The administrators are able to: (1) View and manage the number of registered users and providers. (2) Moderate reviews and comments, to remove inappropriate or fraudulent information. (3) Address user complaints and support tickets.

2.2.3.3.5 Data Analytics and Reports Module

Administrators can generate reports on platform usage. Also, track key metrics such as booking trends, most-searched services, and user demographics to give insights for strategic planning and system improvement.

2.2.4 Non-Functional Requirement

The nonfunctional requirements for the project would go under the FURPS classification, which was conceived by Hewlett-Packard.

FURPS is an acronym that stands for Functionality, Usability, Reliability, Performance, and Supportability. These categories describe the quality attributes and constraints of the system and address how good the system is at performing its functions.

2.2.4.1 Usability

This area refers to the ease of use and user-friendliness of the platform.

2.2.4.1.1 User Interface

The establishment of an intuitive and clean user interface that is consistent throughout the application was ensured via implementation of shadcn/ui and Tailwind CSS. This way, the user interface can afford a clean and satisfactory user experience. The design is accessible to as many users as possible, with the focus on those with lower digital literacy.

2.2.4.1.2 Learnability

New users can navigate the platform seamlessly by searching for a clinic or booking an appointment with minimal guidance.

2.2.4.1.3 Accessibility

The web-application is responsive and functional on web browsers (Google Chrome, Microsoft Edge) and within the WebView-based mobile application on Android devices (8.0 Oreo or higher).

2.2.4.1.4 Feedback

Confirmed appointments receive explicit confirmation messages whereas erroneous form submissions prompt the system into displaying error notifications for such activities.

2.2.4.2 Reliability

This area defines the consistency and dependability of the system.

2.2.4.2.1 Availability

The backend services powering the application are mostly available, with minimal downtime, excluding scheduled maintenance.

2.2.4.2.2 Data Integrity

All data, especially user information, healthcare provider details, and appointment schedules, are accurate and protected from unauthorized modification. The System Administrator is responsible for managing database integrity.

2.2.4.2.3 Recoverability

When system failure happens, there are procedures for restoring the service and

recovering data. The use of a cloud-hosted database like Supabase aids in this by providing robust backend management.

2.2.4.2.4 Fault Tolerance

The system should handle user errors gracefully (e.g., incorrect data entry in forms) and maintain functionality even if minor components fail.

2.2.4.3 Performance

This category specifies the speed and efficiency of the system.

2.2.4.3.1 Response Time

The application must be fast and responsive. Leveraging Inertia.js and React.js that creates a smooth single-page application experience, ensuring that page loads and data filtering feel instantaneous to the user within both the browser and the WebView.

2.2.4.3.2 Load Management

The system is expected to be capable of handling concurrent requests from multiple users without any degradation in performance. Thus, architecture built on Laravel and Supabase is designed to be scalable.

2.2.4.3.3 Real-time Updates

Information such as a provider's operating hours and service availability must be updated and displayed in real-time.

2.2.4.3.4 Resource Consumption

The application is optimized to consume minimal data, battery, and processing power on the client's device, which is crucial for users with limited internet connectivity or older hardware.

2.2.4.4 Supportability

The system is well-maintained and updated regularly.

2.2.4.4.1 Maintainability

The project's code, developed using modern frameworks like Laravel and React with TypeScript, are well-documented, clean, and organized. This allows for easier debugging, updates, and future enhancements.

2.2.4.4.2 Extensibility

The system utilizes an Agile methodology, for easy extensions of new features and modules in the future.

2.2.4.4.3 Testability

The system undergoes unit testing and user acceptance testing to ensure all new features

are working correctly before deployment.

2.2.4.4.4 Installability and Configurability

2.2.4.4.4.1 Installability

The system is accessible in two ways. (1) The primary web-accessible application that requires no installation and can be used directly through a standard web browser. (2) The hybrid mobile application is created by packaging the web application into a native Android WebView container using Android Studio. This generates an APK file for direct download. The installation process requires users to download this file and authorize the installation from an external source on their Android device.

2.2.4.4.2 Configurability

The system is configurable at different user levels. End-users can change their profile settings and notification preferences within the app. Healthcare providers can configure their public profiles, services offered, operating hours, and appointment availability through their dedicated dashboard. The System Administrator has high-level configuration rights over user roles, permissions, and system-wide parameters through a secure admin panel.

2.3 Design

2.3.1 Conceptual Framework

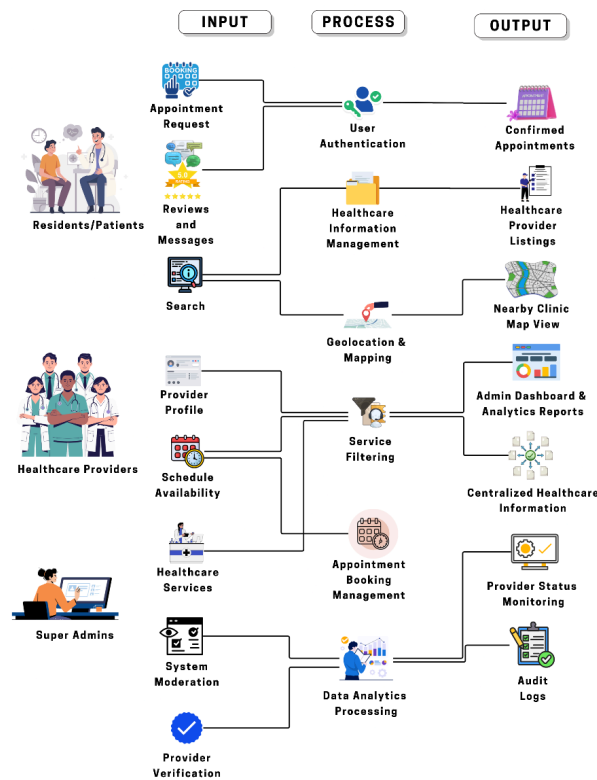


Figure 1. Conceptual Framework

2.3.1.1 Input

The primary inputs to Project HIMSOG are user-account and profile data (resident registrations, provider registrations and verification documents, and Google OAuth credentials), provider-published details (services, operating hours, accepted insurances, prices, photos, and location), user search queries and selected filters (service type, specialty, proximity, availability, insurance), geolocation data (device location or manually entered address), operational updates from providers (availability changes, appointment confirmations/declines, messages), user feedback and ratings

submitted after service, and administrative inputs (provider verification actions, moderation flags, and system configuration parameters).

2.3.1.2 Process

Project HIMSOG checks each user's identity and role before giving access, then stores and retrieves profiles, appointments, and feedback in its database. It allows users to search and filter providers by location and other details, shows maps and routes, and finds the nearest providers. The booking process is step-by-step: check availability, reserve a slot, provider accepts or declines, then sends a notification. It

includes chat for user-provider communication, feedback submission with admin review, and data collection for reports and dashboards. The system is built to be fast, secure, reliable, and recover quickly if there's an issue.

2.3.1.3 Output

The system gives residents ranked search results, maps with directions, confirmed bookings, reminders, booking history, and feedback options. Providers get a dashboard to manage services, schedules, requests, and view booking data. Administrators can review content, verify providers, monitor the system, and generate reports. At the system level, it provides real-time updates, notifications, anonymous data for planning or research, and combined insights to improve services.

2.3.2 Data models

2.3.2.1 Data Flow Diagram

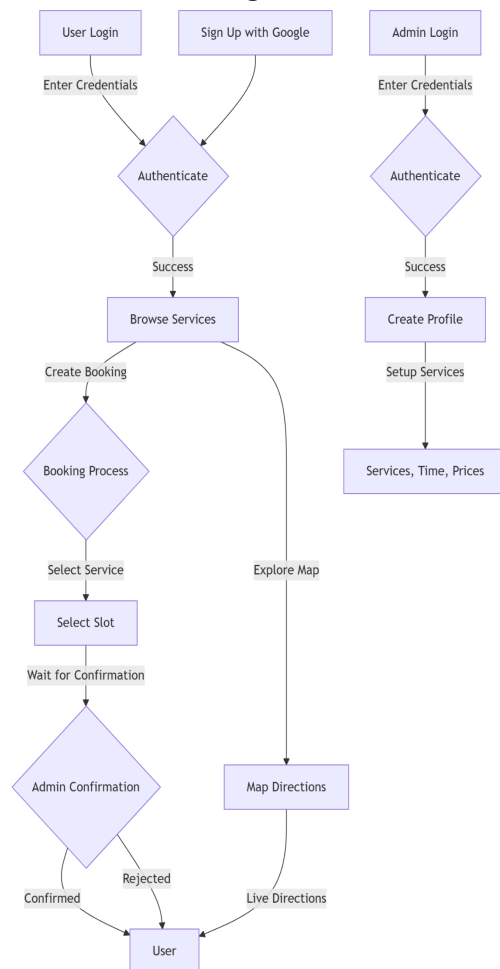


Figure 2: Data Flow Diagram

Figure 2 illustrates the main features and data flow of the Project Himsog platform having

three different users: Admin, Provider and User. When a user logs in, he/she is directed towards a particular dashboard which opens features according to the role. Admins maintain the overall health of the platform, by verifying their providers, moderating the content as well as analyzing the system data. Providers can manage their business details, services, bookings and can reply to reviews. Users can search and book with providers and write reviews, ratings, and manage personal profiles at the same time. These different system components like Database, Map Service, and below Analytics Engine work together to support these various user activities to provide a smooth, interlinked user experience for all stakeholders.

2.3.3 Use Case Diagram

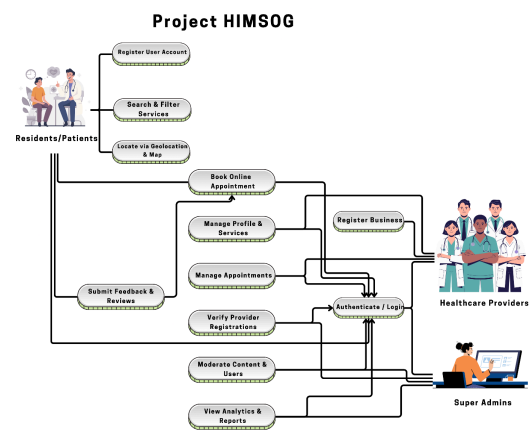


Figure 3: Use Case Diagram

Figure 3 shows the functional capabilities of the Project HIMSOG website and the interactions that each distinct user role can perform. The General User/Resident, who is the primary health service seeker, can do things like register an account, find health-related activities, and book appointments along with providing feedback. Healthcare Providers use the system to manage their profile, their services, as well as their accounts schedule, which they must respond to and manage. The System Administrator acts like the platform manager, making it the topmost user seeing that it has exclusive approval to manage every new health care provider registration and control all user accounts plus all related content (including moderation of reviews) and report generation from the Data Analytics Dashboard to derive insights intended for strategic planning and improvement of the system

2.3.2.2 Entity Relationship Diagram

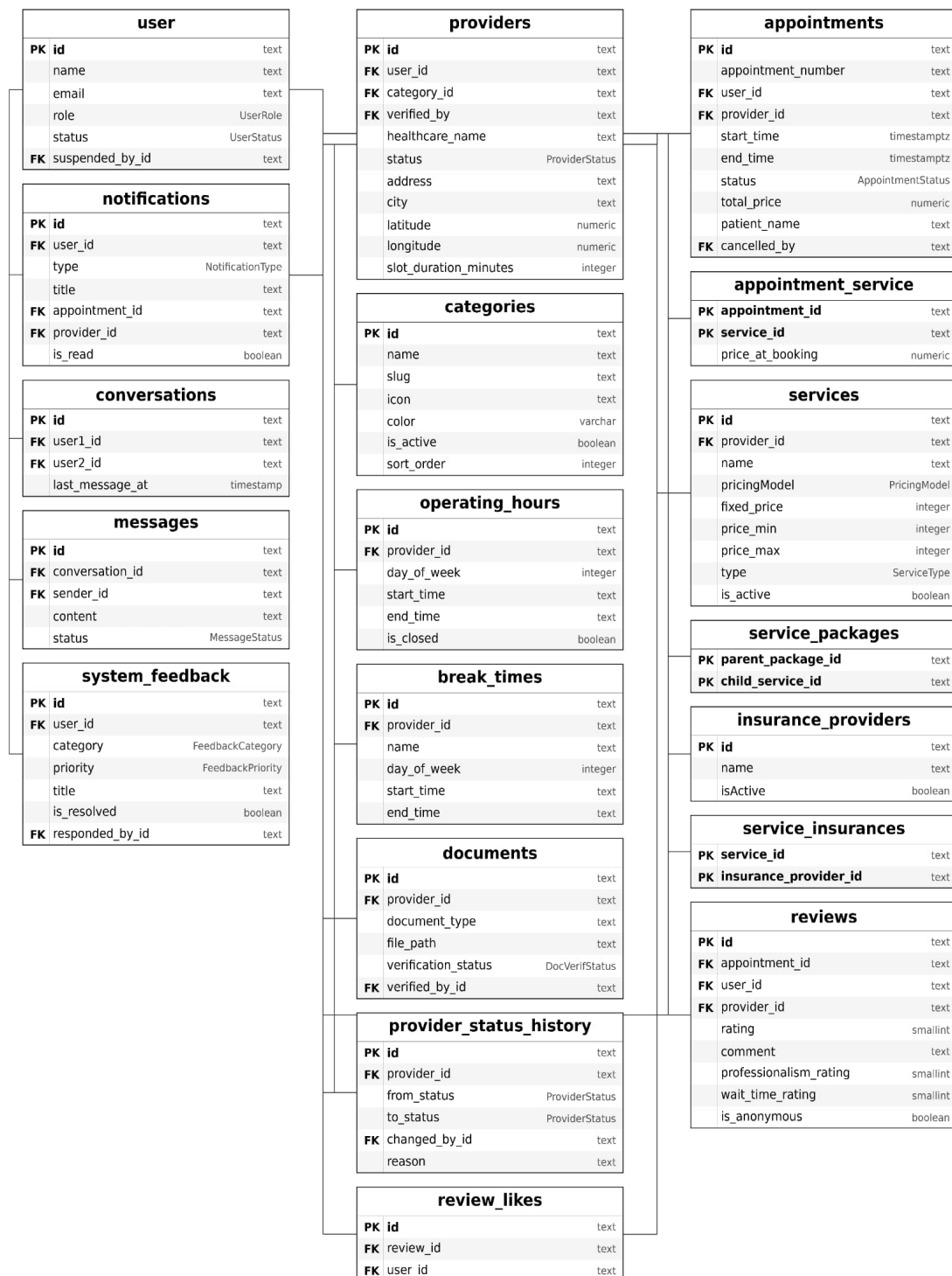


Figure 4: Entity Relationship Diagram

Figure 4 illustrates the Entity Relationship Diagram (ERD) of Project HIMSOG, showing the relationships between the system's core data entities. The diagram includes Users, Healthcare Providers, Services, Appointments, Reviews, and Administrative records. These entities support key system functions such as

2.3.2.3 Data Dictionary

provider registration, service listing, appointment booking, and feedback submission. The relationships between entities ensure proper data organization and enable efficient storage, retrieval, and management of healthcare information within the platform.

TABLE: USER — Core user accounts – patients, providers, admins

#	Field Name	DB Column Name	Data Type	Length / Precision	Constraint	Default	Description
1	id	id	UUID	36	PK, NOT NULL	gen_random_uuid()	Unique user identifier
2	name	name	VARCHAR	255	NOT NULL		Full display name
3	email	email	VARCHAR	255	UNIQUE, NOT NULL		Login email address
4	role	role	ENUM	-	NOT NULL	USER	UserRole: USER ADMIN PROVIDER
5	status	status	ENUM	-	NOT NULL	ACTIVE	UserStatus: ACTIVE SUSPENDED DELETED
6	emailVerified	email_verified	BOOLEAN	-	NOT NULL	false	Whether email has been verified
7	image	image	VARCHAR	500	NULLABLE	NULL	Profile image URL
8	createdAt	created_at	TIMESTAMPTZ	-	NOT NULL	now()	Record creation timestamp
9	updatedAt	updated_at	TIMESTAMPTZ	-	NOT NULL	now()	Last update timestamp
10	suspendedAt	suspended_at	TIMESTAMPTZ	-	NULLABLE	NULL	When the account was suspended
11	suspendReason	suspend_reason	TEXT	-	NULLABLE	NULL	Reason for suspension
12	suspendedById	suspended_by_id	UUID	36	FK → user_id, NULLABLE	NULL	Admin who performed the suspension

Figure 5: Data Dictionary - Users table

TABLE: PROVIDERS — Healthcare provider profiles (one-to-one with a PROVIDER user)

#	Field Name	DB Column Name	Data Type	Length / Precision	Constraint	Default	Description
1	id	id	UUID	36	PK, NOT NULL	gen_random_uuid()	Provider profile ID
2	userId	user_id	UUID	36	FK → user_id, UNIQUE, NOT NULL		Linked user account (1-to-1)
3	categoryId	category_id	UUID	36	FK → categories_id, NULLABLE	NULL	Healthcare category
4	verifiedBy	verified_by	UUID	36	FK → user_id, NULLABLE	NULL	Admin who verified this provider
5	healthcareName	healthcare_name	VARCHAR	255	NOT NULL		Clinic / practice name
6	description	description	TEXT	-	NULLABLE	NULL	Profile description
7	phoneNumber	phone_number	VARCHAR	20	NULLABLE	NULL	Contact phone number
8	email	email	VARCHAR	255	NULLABLE	NULL	Public contact email
9	coverPhoto	cover_photo	VARCHAR	500	NULLABLE	NULL	Cover image URL
10	status	status	ENUM	-	NOT NULL	PENDING	ProviderStatus: PENDING VERIFIED SUSPENDED REJECTED
11	address	address	VARCHAR	500	NOT NULL		Street address
12	city	city	VARCHAR	100	NOT NULL	Digos	City name
13	province	province	VARCHAR	100	NOT NULL	Davao del Sur	Province / state
14	latitude	latitude	DECIMAL	10,8	NULLABLE	NULL	GPS latitude
15	longitude	longitude	DECIMAL	11,8	NULLABLE	NULL	GPS longitude
16	slotDuration	slot_duration_minutes	INTEGER	-	NOT NULL	30	Appointment slot length in minutes
17	verifiedAt	verified_at	TIMESTAMPTZ	-	NULLABLE	NULL	When verified by admin
18	createdAt	created_at	TIMESTAMPTZ	-	NOT NULL	now()	Record creation time
19	updatedAt	updated_at	TIMESTAMPTZ	-	NOT NULL		Last update time

Figure 6: Data Dictionary - Providers

TABLE: SERVICES — Services offered by a provider (single or bundled package)

#	Field Name	DB Column Name	Data Type	Length / Precision	Constraint	Default	Description
1	id	id	UUID	36	PK, NOT NULL	gen_random_uuid()	Service ID
2	providerId	provider_id	UUID	36	FK → providers_id, NOT NULL		Owning provider
3	name	name	VARCHAR	255	NOT NULL		Service name
4	description	description	TEXT	-	NULLABLE	NULL	Detailed description
5	type	type	ENUM	-	NOT NULL	SINGLE	ServiceType: SINGLE PACKAGE
6	pricingModel	pricing_model	ENUM	-	NOT NULL	FIXED	PricingModel: FIXED RANGE INQUIRE
7	fixedPrice	fixed_price	INTEGER	-	NOT NULL	0	Price in minor units (model=FIXED)
8	priceMin	price_min	INTEGER	-	NOT NULL	0	Min price (model=RANGE)
9	priceMax	price_max	INTEGER	-	NOT NULL	0	Max price (model=RANGE)
10	isActive	is_active	BOOLEAN	-	NOT NULL	true	Whether service is available for booking
11	createdAt	created_at	TIMESTAMPTZ	-	NOT NULL	now()	Record creation time
12	updatedAt	updated_at	TIMESTAMPTZ	-	NOT NULL		Last update time

Figure 7: Data Dictionary - Services

TABLE: APPOINTMENTS — Booked appointments between patients and providers

#	Field Name	DB Column Name	Data Type	Length / Precision	Constraint	Default	Description
1	id	id	UUID	36	PK, NOT NULL	gen_random_uuid()	Appointment ID
2	appointmentNumber	appointment_number	VARCHAR	50	UNIQUE, NOT NULL		Human-readable booking reference
3	userId	user_id	UUID	36	FK → user_id, NOT NULL		Patient who booked
4	providerId	provider_id	UUID	36	FK → providers_id, NOT NULL		Provider being seen
5	startTime	start_time	TIMESTAMPTZ	-	NOT NULL		Appointment start (tz-aware)
6	endTime	end_time	TIMESTAMPTZ	-	NOT NULL		Appointment end (tz-aware)
7	status	status	ENUM	-	NOT NULL	PENDING	AppointmentStatus value
8	notes	notes	TEXT	-	NULLABLE	NULL	Patient notes or special requests
9	totalPrice	total_price	DECIMAL	10,2	NOT NULL		Total price at booking time
10	patientName	patient_name	VARCHAR	255	NOT NULL		Patient name captured at booking
11	patientEmail	patient_email	VARCHAR	255	NOT NULL		Patient email captured at booking
12	patientPhone	patient_phone	VARCHAR	30	NULLABLE	NULL	Patient phone captured at booking
13	activityNotes	activity_notes	TEXT	-	NULLABLE	NULL	Provider summary after completion
14	cancelledAt	cancelled_at	TIMESTAMPTZ	-	NULLABLE	NULL	When cancellation occurred
15	cancellationReason	cancellation_reason	TEXT	-	NULLABLE	NULL	Reason for cancellation
16	cancelledBy	cancelled_by	UUID	36	FK → user_id, NULLABLE	NULL	User who cancelled
17	confirmationEmailSentAt	confirmation_email_sent_at	TIMESTAMPTZ	-	NULLABLE	NULL	When confirmation email was sent
18	reminder24hSentAt	reminder_24h_sent_at	TIMESTAMPTZ	-	NULLABLE	NULL	When 24h reminder was sent
19	reminder1hSentAt	reminder_1h_sent_at	TIMESTAMPTZ	-	NULLABLE	NULL	When 1h reminder was sent
20	createdAt	created_at	TIMESTAMPTZ	-	NOT NULL	now()	Record creation time
21	updatedAt	updated_at	TIMESTAMPTZ	-	NOT NULL		Last update time

Figure 8: Data Dictionary - Appointment

#	Field Name	DB Column Name	Data Type	Length / Precision	Constraint	Default	Description
1	id	id	UUID	36	PK, NOT NULL	gen_random_uuid()	Review ID
2	appointmentId	appointment_id	UUID	36	FK → appointments.id, UNIQUE		Source appointment (1-to-1)
3	userId	user_id	UUID	36	FK → user.id, NOT NULL		Review author
4	providerId	provider_id	UUID	36	FK → providers.id, NOT NULL		Provider being reviewed
5	rating	rating	SMALLINT	-	NOT NULL (1-5)		Overall star rating
6	comment	comment	TEXT	-	NULLABLE	NULL	Written review text
7	isAnonymous	is_anonymous	BOOLEAN	-	NOT NULL	false	Hide reviewer identity if true
8	professionalismRating	professionalism_rating	SMALLINT	-	NULLABLE (1-5)	NULL	Professionalism sub-rating
9	cleanlinessRating	cleanliness_rating	SMALLINT	-	NULLABLE (1-5)	NULL	Cleanliness sub-rating
10	waitTimeRating	wait_time_rating	SMALLINT	-	NULLABLE (1-5)	NULL	Wait time sub-rating
11	valueRating	value_rating	SMALLINT	-	NULLABLE (1-5)	NULL	Value for money sub-rating
12	providerResponse	provider_response	TEXT	-	NULLABLE	NULL	Provider's public reply
13	respondedAt	responded_at	TIMESTAMPZ	-	NULLABLE	NULL	When provider responded
14	isEdited	is_edited	BOOLEAN	-	NOT NULL	false	Whether review was edited
15	editedAt	edited_at	TIMESTAMPZ	-	NULLABLE	NULL	Timestamp of last edit
16	createdAt	created_at	TIMESTAMPZ	-	NOT NULL	now()	Review submission time
17	updatedAt	updated_at	TIMESTAMPZ	-	NOT NULL		Last update time

Figure 9: Data Dictionary - Reviews

#	Field Name	DB Column Name	Data Type	Length / Precision	Constraint	Default	Description
1	id	id	UUID	36	PK, NOT NULL	gen_random_uuid()	Log entry ID
2	adminId	admin_id	UUID	36	FK → user.id, NOT NULL		Admin who acted
3	action	action	VARCHAR	100	NOT NULL		Action code (e.g. 'PROVIDER_STATUS_CHANGED')
4	targetType	target_type	VARCHAR	50	NOT NULL		Entity type acted on (e.g. 'Provider')
5	targetId	target_id	UUID	36	NOT NULL		ID of the entity acted on
6	oldValue	old_value	JSONB	-	NULLABLE	NULL	Previous state snapshot
7	newValue	new_value	JSONB	-	NULLABLE	NULL	New state snapshot
8	metadata	metadata	JSONB	-	NULLABLE	NULL	Additional context
9	ipAddress	ip_address	VARCHAR	45	NULLABLE	NULL	Admin IP address
10	userAgent	user_agent	TEXT	-	NULLABLE	NULL	Admin browser user-agent
11	createdAt	created_at	TIMESTAMPZ	-	NOT NULL	now()	Timestamp of action

Figure 10: Data Dictionary - Audit Logs

Figure 5 illustrates the user table that stores the account information of individuals who access the system, including residents, healthcare providers, and administrators. It contains essential user details such as identification, login credentials, and role information used for authentication and system access control.

Figure 6 illustrates the providers table that contains the information of registered healthcare providers in the system. It includes details such as facility name, location, contact information, services offered, and other operational data necessary for users to identify and connect with healthcare facilities.

Figure 7 illustrates the services table that records the different healthcare services offered by registered providers. It stores information such as service name, description, price range, and the associated provider, allowing users to search and filter healthcare services efficiently.

Figure 8 illustrates the appointments table that manages the booking records between users and healthcare providers. It includes details such as appointment date, time, service selected, booking status, and the associated user and provider.

Figure 9 illustrates the reviews table that stores feedback and ratings submitted by users after receiving healthcare services. This information helps other users evaluate healthcare providers and supports transparency in service quality.

Figure 10 illustrates the audit logs table that records important actions performed within the system by users, healthcare providers, and administrators. It stores information about activities such as account updates, provider verification, appointment changes, and other system events.

2.3.6 Architecture Design

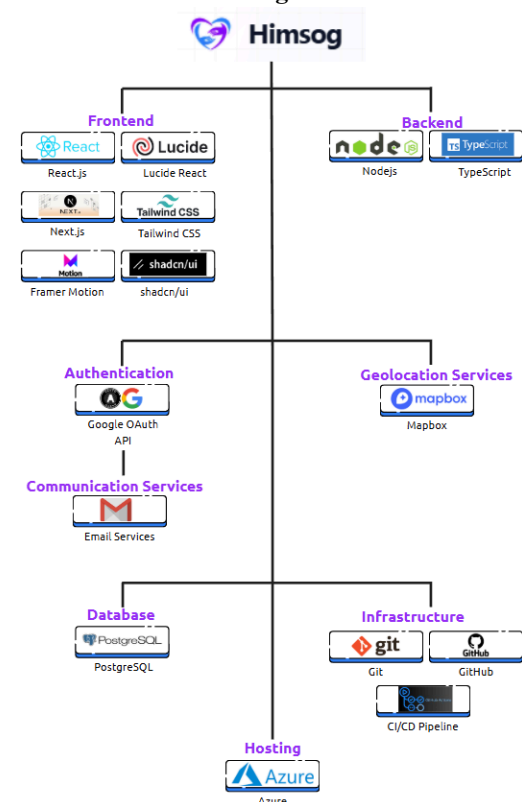


Figure 11: System Architecture

The system architecture for Project Himsog is illustrated in Figure 4. It shows how the main components of the Project Himsog application are designed to work together. The frontend of the application was developed using React.js and Next.js, as well as other libraries that allow the web application interface to be responsive. The backend (Node.js/TypeScript application) deals with the application business logic and allows for communication between the components in the system. All user authentication is completed using Google OAuth and Mapbox provides both mapping and geolocation capability for the system. The Project Himsog application's primary database is PostgreSQL and is used for storing the structured application data. The Project Himsog application uses Git with GitHub and a CI/CD pipeline to manage version control and releases, while the application is hosted on Microsoft Azure and is highly scalable and available.

2.4 Development

2.4.1 Software Process Model

Project HIMSOG stands ideal under Agile methodology; user-centric, adaptive, and iterative [20]. The project cuts across designing accessibility in healthcare for the residents of Digos City, which needs input and re-adjustment from user feedback to see if that effective ever truly becomes useful and user-friendly to its target audience [21]. In addition, Agile allows the team to build the system through sprints or incremental releases, repeating phases, ensuring a continuous loop of progress and refinement. [23].

Furthermore, Agile framework reduces risk as well as fosters improved collaboration. By developing and testing different features in shorter cycles. This collaborative and transparent process ensures that Project HIMSOG is not just executed smoothly but rather continues evolving over time to sustain the community's needs, making it a stronger, more effective, and more impactful application overall [20].

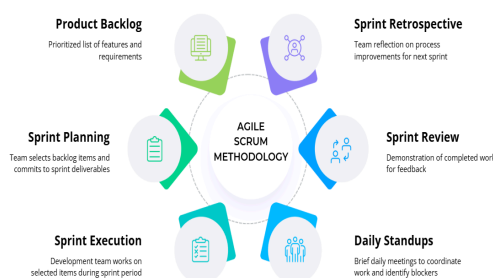


Figure 12: Agile Scrum Methodology

Figure 12 presents the Agile Scrum methodology adopted in the development of Project HIMSOG. The researchers selected Scrum to enable an iterative and flexible approach, allowing continuous improvement of system features based on user feedback. The process begins with a prioritized Product Backlog, followed by Sprint Planning, where tasks are selected for development. During Sprint Execution, the team develops the selected features while conducting Daily Standups for progress tracking. Each sprint concludes with a Sprint Review to gather feedback and a Sprint Retrospective to identify improvements for the next cycle.

2.4.2 Project Technology

2.4.2.1 React .js

One of the most famous frontend JavaScript libraries that aims to help you create an interactive or reactive user interface. This language was chosen for frontend development because it makes it easier to build complex UI because of its declarative way of defining the desirable final states of the UI [24]. This is used to build the core user interface—including the clinic directory, appointment forms, filtering menus, and user dashboards. Its component-based structure allows the system to reuse UI elements and update pages instantly without reloading, improving overall performance and user experience.

2.4.2.2 Typescript

Is a multi-faceted tool: a programming language, a type checker, a compiler, and a very powerful language service provider. Typescript is, in fact, a superset of JavaScript. Typescript is adding static typing ability to a developer, enabling them to define data structures and impose constraints on the way that data is used within the application, allowing its type checker to catch some types of errors, providing freedom through restriction by avoiding common bugs, and compiling that to normal JavaScript. Also, adds all special type services by providing features like intelligent code completion and real-time error feedback. Using TypeScript in Project Himsog makes a strong and maintainable front-end for data consistency for features that are quite complex like appointment booking and filtering and accelerates development [25].

2.4.2.3 Next.js

Next.js is a modern and highly efficient JavaScript framework designed for building fast, scalable, and user-friendly web

applications. Next.js functions as the primary frontend framework that manages page rendering, handles client-side routing, processes user interactions, and communicates with backend APIs for tasks such as authentication, appointment scheduling, and provider data retrieval. It serves as the main interface layer, efficiently processing user requests and providing dynamic, responsive pages for the system.

2.4.2.4 Resend

A modern and developer-friendly email delivery platform designed for sending transactional and notification emails with high reliability. In this project, Resend is used to manage all email-based communication between users and healthcare providers, particularly in sending inquiry messages, account-related notifications, and system alerts. Its integration helps maintain efficient communication while reducing backend complexity.

2.4.2.5 Tailwind CSS

A utility-first CSS framework for rapidly building a custom user interface. Instead of writing custom CSS, you apply existing utility classes directly in HTML to style elements. This approach promotes a consistent design throughout the system, faster development time, and a more maintainable codebase. This enabled a quick development without the inertia of highly-considered design principles to generate a neat and satisfactory user experience [27].

2.4.2.6 Framer Motion

This library provides React applications with an open-source motion library to allow developers to achieve the greatest complexity, smoothness, and interactivity in their animations with the least amount of code. It covers the important parts of layout transitions, gestures, and shared element animations that create better user experience. It enhances the visual experience through smooth page transitions, animated models, loading indicators, and interactive map animations—making navigation feel modern and fluid.

2.4.2.7 shadcn/ui

A collection of beautifully designed and reusable UI components to copy directly from source code instead of installing a traditional library in a developer's project, built over the top of Radix UI for accessibility, and styled with Tailwind CSS. This allows developers to fully customize every component to the precise

specifications: in this case, shadcn/ui was utilized in the system as a base to quickly build a pretty uniformized, accessible front-end interface, including all dashboards, forms, and other interactive parts. In Project Himsog, shadcn/ui provides the foundation for components such as buttons, dropdowns, dialogs, tables, and input forms, ensuring accessibility and consistent styling throughout the application.

2.4.2.8 PostgreSQL

PostgreSQL is a powerful, open-source relational database management system known for its reliability, scalability, and strong support for complex queries and data integrity. In Project Himsog, PostgreSQL is used as the primary database for storing structured data such as user accounts, healthcare provider profiles, verification documents metadata, appointment records, reviews, and system logs. Its support for relational constraints and transactional consistency ensures accurate data handling and secure storage, making it suitable for managing the platform's core operational data.

2.4.2.9 Mapbox

Mapbox is a location platform company supplying tools for visualizing and analyzing spatial data. Mapbox ranks among the largest commercial providers of high-performance maps and navigation tools, serving enterprises, government agencies, and organizations of all sizes [29]. powers the map-based features, including rendering healthcare facility locations, showing user positions, providing navigation routes, and integrating with the KNN algorithm for nearest facility recommendations.

2.4.2.10 Google OAuth API

Through the Google OAuth protocol, user authentication is possible with great security and reliability. This allows any application to verify a user's identity without actually having to deal with the user's passwords. This helps users sign up and log into a service using their Google account thereby making the whole process swift [30].

2.4.2.11 Git

During the software development process, developers track changes in the source code known as distributed version control system/Git. This allows several people to work at the same time on a single project and keep an entire history of changes such that the older versions can be reverted back to when required. Its branch and merge functionalities enable effective updating and parallel

development workflows to be managed. In this system, Git is used to organize the version management of all source codes during development [31].

2.4.2.12 Github

GitHub is a platform that has been developed around Git which is a cloud-based application, providing software development projects with hosting and collaboration, version control, as well as issue-tracking tools. Github also allows people to work as teams to store their repositories online and manage their pull requests to review code changes [32]. Github also supports many different tools and services for software development, thus becoming the main hub for open-source as well as private projects. In this system, GitHub was utilized as the primary repository hosting service as well as collaborative workspace for the development team.

2.4.2.13 K-Nearest Neighbors Algorithm (KNN)

The K-Nearest Neighbors algorithm is considered an instance-based, non-parametric machine learning algorithm for classification and regression. This algorithm predicts by the most common label of the 'K' closest data points from a query point or by averaging their value for regression. KNN works to recommend the nearest healthcare facility based on the user's geolocation, so all recommendations stay relevant and accessible [33]. KNN is used to compute the closest clinics using stored latitude and longitude coordinates. The algorithm ranks facilities by proximity and displays the nearest options on the map and in the directory list, helping users quickly find accessible healthcare services.

2.4.2.14 Lucide React

Lucide React is an open-source icon library that provides a wide collection of clean, customizable, and lightweight SVG icons for React applications. Lucide React supplies consistent visual icons across the platform, enhancing user navigation and improving the clarity of UI elements such as buttons, indicators, menus, and interactive components. The library's flexibility and ease of integration contribute to a more intuitive and visually appealing user experience.

2.4.2.15 Prisma ORM

Prisma ORM is a modern, type-safe Object-Relational Mapping tool designed to simplify database interaction in full-stack applications. Prisma ORM provides an intuitive schema-based approach for modeling

data, automating migrations, and ensuring reliable communication between the application and databases such as PostgreSQL used to manage all interactions with the healthcare database, including storing provider details, handling user appointments, retrieving clinic information, and processing user inquiries. Its structured workflow ensures consistent data accuracy and supports the scalable backend architecture required for the platform.

2.4.2.16 Node.js

Node.js is a server-side JavaScript runtime used to implement the backend of Project Himsog. Node.js handles API requests, authentication workflows, appointment processing, provider verification logic, and communication between the frontend and the PostgreSQL database. Its event-driven and non-blocking architecture supports efficient handling of concurrent user requests and real-time system operations.

2.4.2.17 Azure

Azure serves as the cloud hosting platform for Project HIMSOG. The platform is utilized to deploy and manage the application's frontend, backend services, and database infrastructure, ensuring scalability, high availability, and secure cloud-based deployment. Azure also provides integrated services for data storage, authentication, and monitoring, enabling efficient system management and performance tracking. The platform supports reliable access for users across different devices and locations, contributing to consistent system operation and overall service stability.

2.5 Testing

2.5.1 Unit Testing

Unit testing was a verification process aimed at testing individual modules of the system for correctness and reliability. Each function of the module was subjected to independent testing to confirm that it executed the task correctly before it was integrated into the complete system. This allowed identification and fix of logic errors, mistakes in user interface, and data handling issues. Making sure that each part of the system worked well at this point helped keep the system stable and made it easier for integration during later testing stages.

2.5.2 Alpha Testing

The team conducted the alpha test stage of Project Himsog under the supervision of the capstone project advisor. The system underwent an evaluation before its deployment

to identify any glitches, functionality issues, and unsupported functionality within the system. The evaluation of the system especially aimed to verify the basic functionality of the system on the criteria of test cases that meet the requirements of the proposed functionality of the project. These aspects of functionality considered in the evaluation of the system include searching/filtering of healthcare amenities, displaying data of healthcare providers, displaying data of services, asking queries, and analyzing responses of the system, along with the functionality of the mobile system. A properly designed evaluation checklist is used to measure the correctness of the expected output of the system.

2.5.3 Beta Testing

Beta testing was conducted with the panelists and selected healthcare providers in Digos City to evaluate Project HIMSOG under real-world conditions. The system was tested on both web browsers and the Android mobile application, focusing on key functionalities such as searching for healthcare services, utilizing geolocation and map navigation, viewing provider information, and booking appointments. The testing phase also assessed system performance, response time, and compatibility across different devices and network conditions. Participants were asked to perform specific tasks to measure ease of use, navigation flow, and overall user satisfaction. Feedback from the participants helped identify minor usability issues and interface improvements, which were addressed prior to final deployment. Overall, the beta testing phase confirmed the system's functionality, reliability, usability, and readiness for implementation.

2.5.4 User Acceptance Testing

The User Acceptance Testing for the Project Himsog was executed with the aim of ensuring that the system was functional, and relevant according to the needs of the actual users of the system in the Digos City setting. As the main processes of the system were accomplished, the team engaged with private health facilities in Digos City. The users were instructed on how to use the main functionalities of the system, which include searching health services and viewing health clinic information as well as how to raise queries using the system. The findings from the process of the user acceptance test confirmed that the intended objectives for the development of the Project Himsog system were accomplished.

III. RESULTS AND DISCUSSIONS

3.1 Implementation Plan

Table 4: Implementation Plan

Strategy	Activities	Person(s) Involved	Duration
Information Distribution	System Overview and Website Presentation	Researchers, End-Users (Clinic Staff)	1 hour
System Access Setup	Website Access Configuration and Account Assistance	Researchers, End-Users	20 minutes
Training	Hands-on System Demonstration and Guided Usage	Researchers, End-Users	1 hour

3.2 Implementation Results

With the implementation of Project HIMSOG, which illustrates how the system can fulfill the objectives of the study based on the proposed system model. Based on the results, the system can satisfy the requirements, considering it provides a centralized way of managing healthcare information in Digos City, aside from functioning effectively in support of the goals of the study for both desktop and mobile platforms via the system's main functionalities of healthcare search, filtering, and location services in Project HIMSOG.

3.2.1 To design and implement a digital platform that integrates and consolidates all essential healthcare information such as clinic services, free barangay health services, operating hours, accepted insurance, and contact details, into a single portal for easy access for the city of Digos residents.

After the implementation, it has been observed that the first objective of the study has been successfully achieved. Project HIMSOG has successfully offered a digital platform that combines all relevant health care information from scattered sources, including the services offered from the clinic, the free health care service offered from the barangay government, the operational hours and accepted health care insurance.

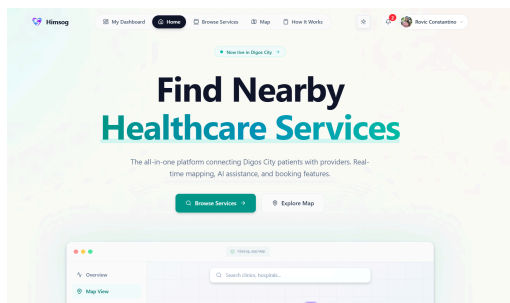


Figure 13: Home Page

Fig. 13 shows the HIMSOG Home Page, which is actually the main gateway of the application for the user to interact with. Through the home page, the search feature is accommodated to give Digos City residents an avenue to readily search for healthcare providers. The home page offers a unified layout to enable the user to assess the health services provided without having to search from several distant sources.

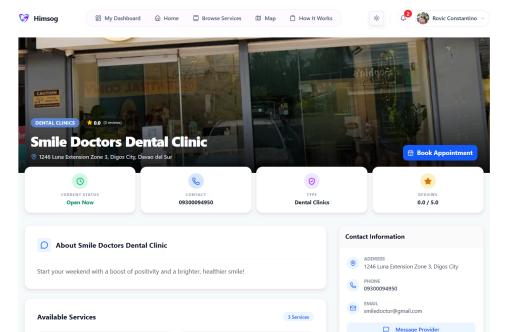


Figure 14: Clinics Profile Page

Figure 14 highlights the Clinic Profile Page of the HIMSOG system. The Clinic Profile Page presents comprehensive information associated with a selected private clinic, including offered services, operating hours, accepted insurance providers, contact details, and clinic location. The page also integrates map-based visualization to assist users in identifying the exact location and planning visits efficiently. Organized layout and clear presentation of information improve readability and user navigation across different sections. Providing all essential clinic details within a single interface enhances decision-making, reduces search time, and improves accessibility to healthcare services.

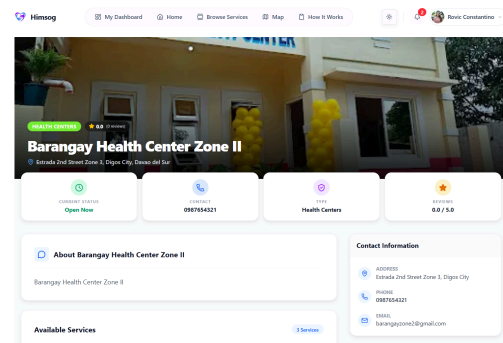


Figure 15: Health Center Profile Page

Figure 15 shows the Health Center Profile Page. This is a display feature that provides information on health centers offered by the government and free health services offered in barangays. This contains offered services, operating hours, and contacts that allow easy identification and availability of public healthcare resources for communities to freely access them through this system.

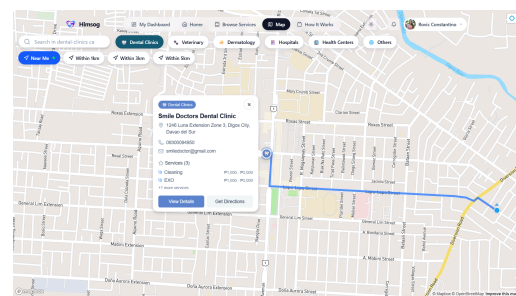


Figure 16: Map Feature

Figure 16 above is the map feature for the project. The map feature gives the user the opportunity to locate the nearest health providers based on the user's location and the health providers in the regions the user has chosen. The map feature uses markers for clinics and health centers that the user can click and read some basic information such as the names and health services offered.

3.2.2 To utilize geolocation and mapping technologies that enable users to conveniently identify and navigate to nearby healthcare providers, improving overall access to services based on proximity and user needs.

After implementation, the second objective has been met. Project HIMSOG integrates geolocation capabilities and mapping functionality to assist a user in identifying nearby healthcare facilities.

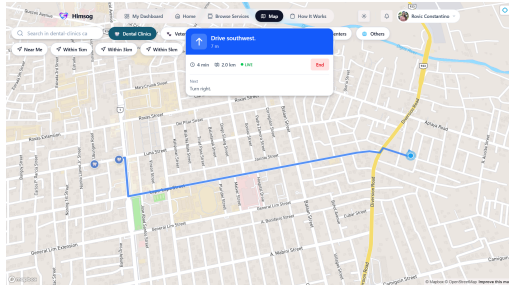


Figure 17: Map Navigation

Figure 17 above shows the Map Navigation feature offered by the HIMSOG system. This feature helps users to navigate from their current location to the preferred healthcare provider. The system helps users in their navigation to healthcare facilities through the display of the path to be followed to reach healthcare facilities and thus improves convenience and helps in quick access to medical services.

3.2.3 To integrate descriptive data analytics that enables administrators to monitor platform usage with respect to booking history (daily, monthly) and often searched services thus deriving insights into trends in healthcare services, demand for service providers, which facilitates data-driven planning, resource allocation, and strategy formulation.

After the implementation of the project, the third objective of the project has been satisfied. The project HIMSOG constitutes an effective integration of a descriptive data analytics feature that enables the administrators to monitor the service usage of the platform regarding booking history and also the most frequently booked services. The service feature provides information regarding service trends and demand.

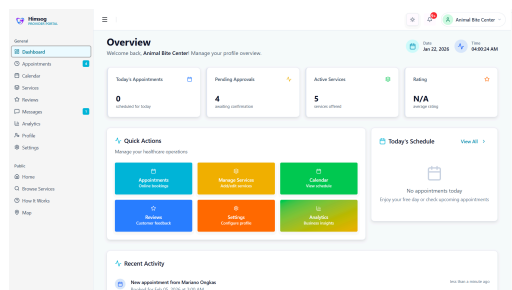


Figure 18: Healthcare Provider Portal -Dashboard

Figure 18 above shows the Healthcare Provider Portal Dashboard of Project

HIMSOG. The portal offers healthcare providers an insight into their operations within the platform by offering access to appointment notifications, services, and message notifications. It is considered the control center of operations that helps healthcare providers monitor their operations with users.

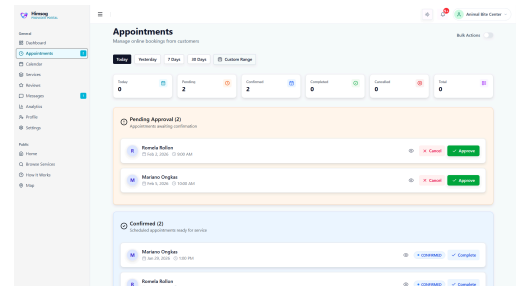


Figure 19: Healthcare Provider Portal - Appointment Dashboard

Figure 19 illustrates the Appointments Dashboard within the Healthcare Provider Portal. This screen allows providers to view, confirm, or manage appointment requests submitted by users. The dashboard organizes appointments based on status, enabling providers to track daily and upcoming bookings, which supports efficient scheduling and service delivery.

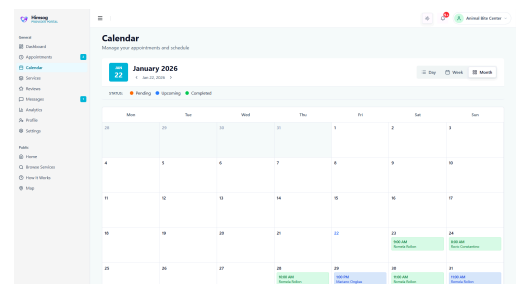


Figure 20: Healthcare Provider Portal - Calendar View

Figure 20 illustrates the Calendar View of the Healthcare Provider Portal. This feature presents appointments in a date-based format, allowing providers to visualize their schedules over time. The calendar view helps healthcare providers monitor daily and monthly appointment distributions, supporting better planning and time management.

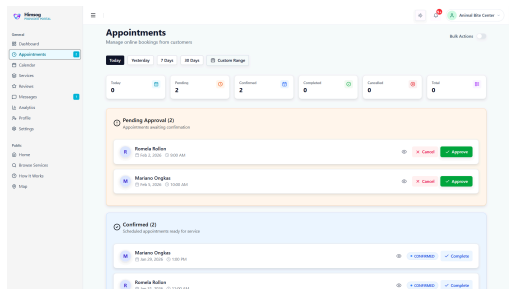


Figure 21: Healthcare Provider Portal - Booking History Analytic

Figure 21 illustrates the Booking History Analytics feature of the Healthcare Provider Portal. This analytic view displays historical booking data, enabling providers to monitor trends in appointment volume over specific periods such as daily and monthly intervals. The availability of booking history supports data-driven insights into service demand and operational performance.

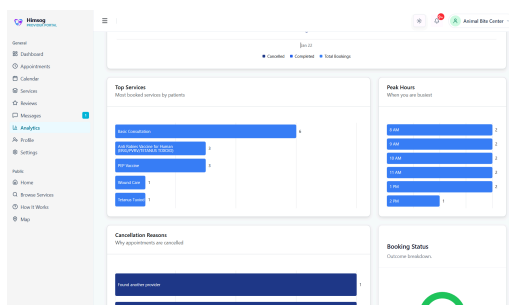


Figure 22: Healthcare Provider Portal - Frequently Booked Services Analytic

Figure 22 illustrates the Frequently Booked Services Analytics feature. This screen highlights the most commonly booked healthcare services on the platform, allowing providers to identify service demand patterns. These insights assist healthcare providers in optimizing service offerings and improving resource allocation based on user demand.

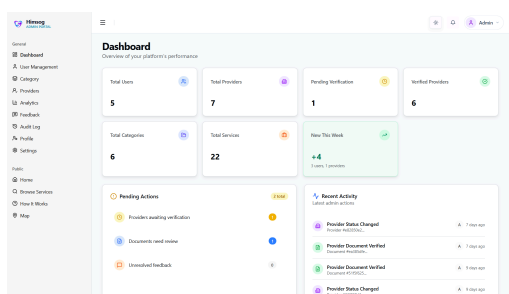


Figure 23: Super Admin Portal - Dashboard

Figure 23 illustrates the Super Admin Portal Dashboard of Project HIMSOG. This dashboard provides administrators with a comprehensive overview of platform-wide activity, including registered users and healthcare providers. It enables administrators to monitor system usage and maintain operational oversight.

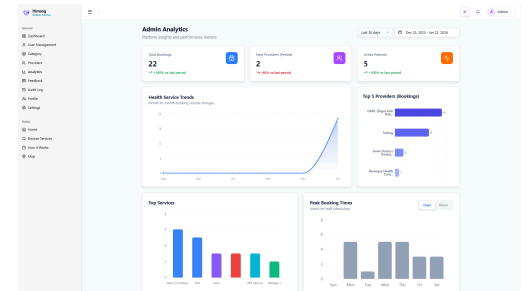


Figure 24: Super Admin Portal - Analytic

Figure 24 illustrates the Analytics module within the Super Admin Portal. This feature aggregates platform usage data, including booking trends and frequently accessed services. The analytics module supports administrators in deriving insights for strategic planning, resource allocation, and policy formulation aimed at improving healthcare service delivery in Digos City.

IV. CONCLUSION AND RECOMMENDATION

4.1 Conclusion

Based on the results of the study, the researchers concluded that Project Himsog was successfully developed and implemented in accordance with the defined objectives. The system effectively provides a centralized web-based platform that connects users with nearby healthcare service providers through secure authentication, geolocation services, and appointment management. Furthermore, the study yielded the following findings:

4.1.1 The integration of geolocation and mapping services has significantly improved the accessibility of healthcare services by enabling users to locate nearby providers, view routes, and make informed decisions based on location and availability. This feature transformed a traditionally manual search process into a more efficient and user-centered experience.

4.1.2 The implementation of a role-based system for users, providers, and administrators has streamlined system operations and data

management. Providers are able to manage their service offerings and bookings, while administrators can oversee verification and moderation processes, resulting in improved platform reliability and operational control.

4.1.3 The use of modern web technologies such as React, Node.js, PostgreSQL, and cloud hosting ensured system scalability, responsiveness, and security. These technologies contributed to a stable platform capable of handling real-time interactions, data storage, and external service integrations, thereby enhancing overall system performance and user satisfaction.

4.2 Recommendation

Based on the findings of the study, the researchers propose the following recommendations for future researchers and developers to further enhance Project Himsog:

4.2.1 Advanced Analytics and Reporting

It is recommended that future versions of the system incorporate advanced analytics tools to track booking trends, service demand, and user behavior. These insights could support administrators and providers in making data-driven decisions and improving service delivery.

4.2.2 Integration with Additional Healthcare Services

Future researchers may consider integrating Project Himsog with other healthcare-related systems such as electronic medical records, pharmacy services, or insurance verification platforms. This would further expand the system's functionality and contribute to a more comprehensive digital healthcare ecosystem.

4.2.3 Automated Provider Document Verification

The researchers recommend the integration of an automated document verification system to support the provider registration and approval process. Currently, healthcare providers are required to submit official documents to prove their legitimacy, which are manually reviewed by the super administrator. Future enhancements may incorporate AI-assisted document validation, such as Optical Character Recognition (OCR), document classification, and cross-checking with Philippine government registries (e.g., PRC, DOH, or LGU-issued permits).

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